Experiment - 12

Aim -

To implement echo client server using TCP/IP sockets and a chat program.

TCP

```
import socket
import threading

Server.
def start_server():
    server = socket.socket(socket.AF_INET,
                           socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    conn, addr = server.accept()
    print(f"[SERVER] Connected by {addr}")
```

Client

```
def start_client():
    client = socket.socket(socket.AF_INET,
                           socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")

# main
if __name__ == "__main__":
    server_thread = threading.Thread(target = start_server)
    server_thread.start()
    start_client()
```

Output:

```
[SERVER] Listening on 127.0.0.1:5000 [client]
[SERVER] Connected by ('127.0.0.1', 53458) Enter message (or 'quit' to exit):
Hello
[SERVER] Received: HELLO
[SERVER] Response server received: Hello
Enter message (or 'quit' to exit): quit
[SERVER] Connection closed by client.
```

UDP:-

```
import socket
import threading

# -- Server --
def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(("127.0.0.1", 12000))
    print("[server] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[server] client {addr}")
    s.close()
    print("[server] stopped.")

# -- Client --
def udp_client():
    c = socket.socket(socket.AF_INET, socket.
                      SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
```

```
c.sendto(msg.encode(), serv)
if msg.lower() == 'quit':
    print("[client] closing connections...") break
    data, _ = conect.from(1024)
    print("Echo from server:", data.decode())

# main --
if __name__ == "__main__":
    threading.Thread(target = udp_serv, daemon
                                    = True)
    start().udp_client()
```

Output:

[Server] Listening on 127.0.0.1:5000 [client] connected
[Server] Reciever : Hello
Enter message (or 'quit' to exit) : quit
[Server] Connection stopped

Result:

Thus, the program was successfully executed.